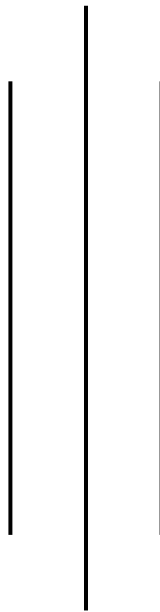




Aircraft Manufacturing Lab Report : Grumman F3F Biplane Model



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Acknowledgment

We would like to extend our heartfelt gratitude to the **Department of Mechanical and Aerospace Engineering** for providing us with the essential resources and equipment required to successfully complete this project. The guidance and state-of-the-art facilities made available to us were instrumental in enhancing our understanding of the fabrication and design processes of a **Grumman F3F Biplane** structural component.

We are deeply grateful to **Professor Arun Bikram Thapa** for assigning this group project and for his unwavering support and encouragement throughout the duration of this lab work. His insightful guidance and mentorship have significantly enriched our learning experience, enabling us to grasp fundamental concepts in aircraft manufacturing and structural design.

Finally, we would like to acknowledge the unwavering dedication and collaborative efforts of our team members. Their hard work, commitment, and teamwork were pivotal in the successful completion of this project.

1 Introduction

This report details the design, fabrication, and testing processes employed to create a scale model of the Grumman F3F biplane. The Grumman F3F was the last biplane fighter aircraft delivered to the United States Navy, characterized by its distinctive barrel-shaped fuselage, radial engine, and double-wing configuration. Our team selected this historically significant aircraft for its unique aerodynamic challenges and manufacturing complexity. We implemented a 1:10 scale model of the original aircraft, maintaining its design characteristics while adapting the structure for model construction. This report outlines our approach to preliminary design, detailed design analysis, material selection, manufacturing processes, and testing procedures.



Figure 1: Grumman F3F

2 Preliminary Design

The initial design phase began with scaling the full-size Grumman F3F by a ratio of 1:10. All dimensions from the original aircraft were proportionally reduced as follows:

2.1 Scaling Considerations

Dimensions	Full-Scale	1:10 Scale
Length	7.32 m	0.73 m (73 cm)
Upper Wingspan	9.75 m	0.98 m (98 cm)
Lower Wingspan	9.14 m	0.91 m (91 cm)
Height	3.02 m	0.30 m (30 cm)
Wing Area	24.15 m ²	0.24 m ²

Table 1: Scaling dimensions from full-scale to model scale

During scaling, special attention was given to the biplane configuration, which required maintaining the original stagger (the horizontal offset between the upper and lower wings) and the gap (the vertical distance between wings). The characteristic barrel-shaped fuselage and rounded engine cowlings were carefully preserved in the scale model to maintain the authentic appearance and aerodynamic properties.

3 Design Considerations

3.1 Materials Selection

- **Plywood:** Used for the fuselage, wing spars, and parts of the landing gear

3.2 Structural and Aerodynamic Design

- **Wing Loading Calculation:** Assuming a model weight of 175kg and a wing area of 0.127 m², the estimated wing load is 1374.464 kg / m².
- **Airfoil Selection:** NACA 2214 for the upper wing and a slightly modified NACA 2214 with reduced camber for the lower wing, compatible with the original F3F's aerodynamic profile.
- **Dihedral Angle:** 2° on the lower wing, 0° on the upper wing, consistent with the original design.
- **Biplane Configuration:** The upper wing is positioned forward of the lower wing (positive stagger) with N-struts and cross bracing to ensure structural rigidity.
- **CG (Center of Gravity):** Positioned approximately 25% of the Mean Aerodynamic Chord (MAC) of the lower wing.

3.3 Project Planning

The project was organized according to the following Gantt chart timeline:

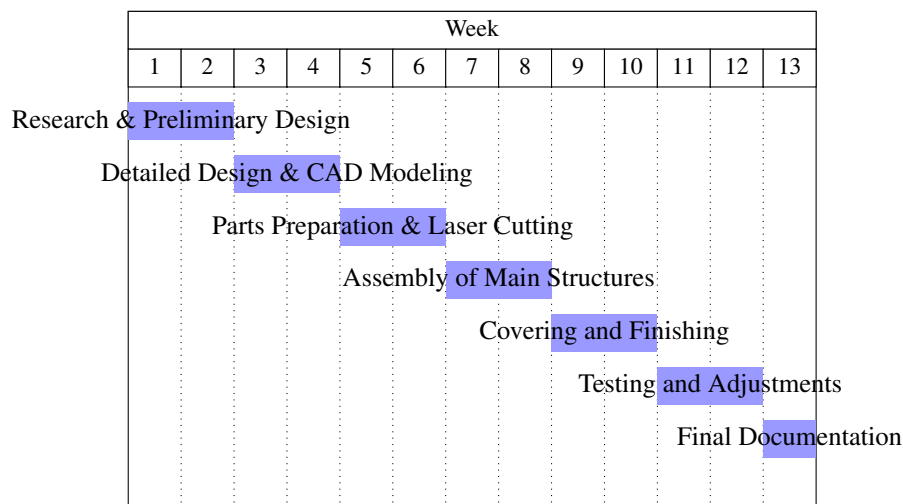


Figure 2: Gantt chart for project planning

4 Detail Design

The design process involved extensive use of XFLR5 software for aerodynamic analysis and CATIA for 3D modeling. The biplane configuration presented unique challenges, particularly in modeling the interference effects between the upper and lower wings.

4.1 Wing Design

The biplane wing configuration required careful analysis of the aerodynamic interaction between the upper and lower wings. Wing ribs were designed with lightening holes to reduce weight while maintaining structural integrity. The NACA 2214 airfoil was selected for its well-documented performance characteristics at low Reynolds numbers and its relatively flat bottom surface, which simplifies construction.

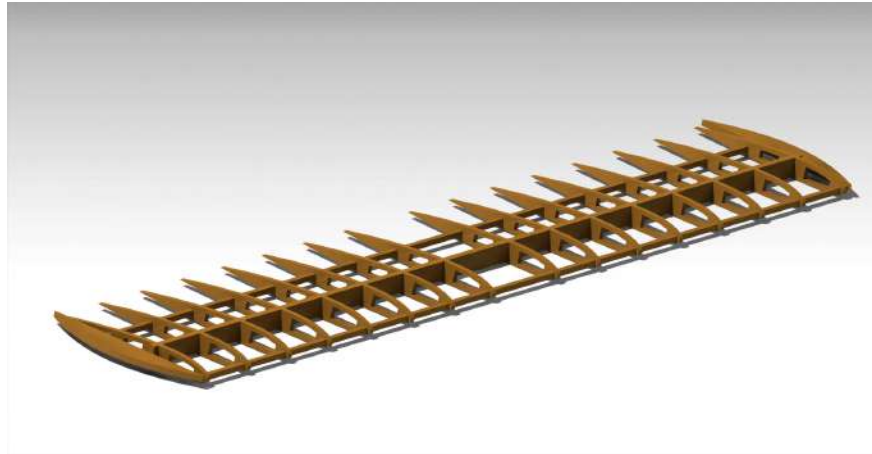


Figure 3: Wing Design in CATIA V5

4.2 Fuselage Design

The distinctive barrel-shaped fuselage of the F3F was recreated using a series of formers and stringers. The cockpit area includes a scale instrument panel and pilot seat. The engine cowling, designed to house a radial engine on the original aircraft, was replicated in detail with cooling vents and a circular front opening.

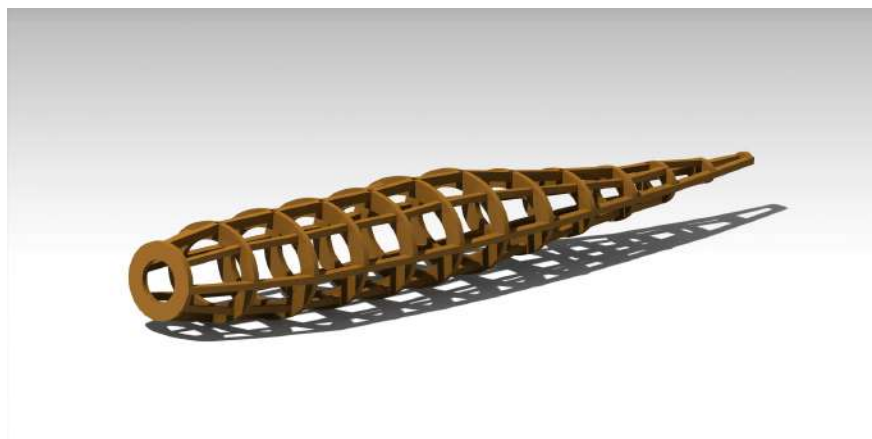


Figure 4: Fuselage Design in CATIA V5

4.3 Control Surfaces

The control surfaces (ailerons, elevator, and rudder) were not included in the manufacturing process. The horizontal and vertical stabilizer was designed.



Figure 5: Vertical Stabilizer

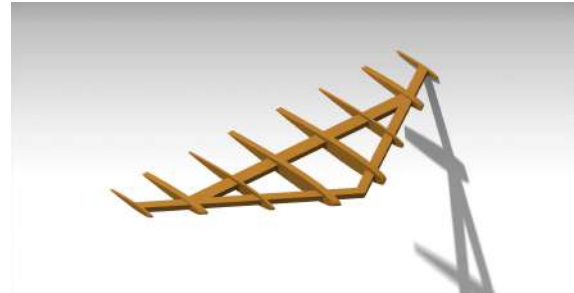


Figure 6: Horizontal Stabilizer

4.4 Assemble

All design components of the Grumman F3F were assembled in CATIA V5.

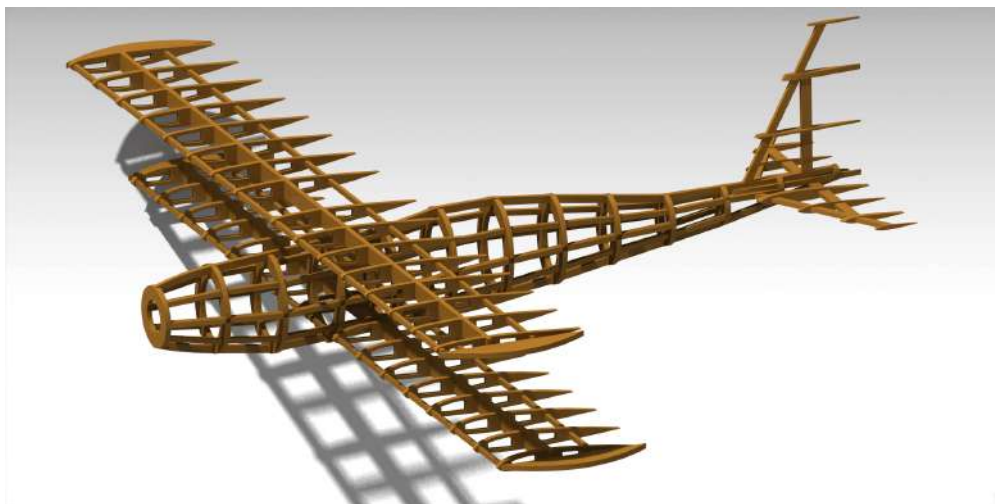


Figure 7: Final Design of Grumman F3F in CATIA V5

4.5 Simulation and Analysis

Computational analysis in XFLR5 confirmed that our scale model would have satisfactory stability characteristics. We ran simulations at various angles of attack and sideslip angles to ensure the model would handle predictably in flight.

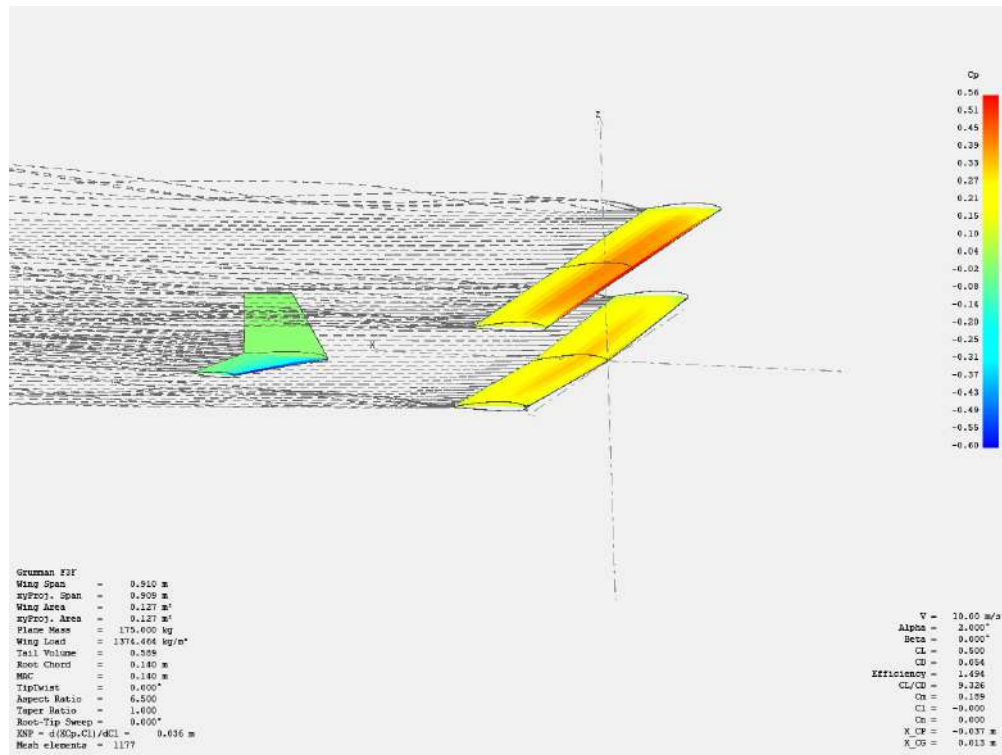


Figure 8: Aerodynamic analysis of Grumman F3F on xflr5

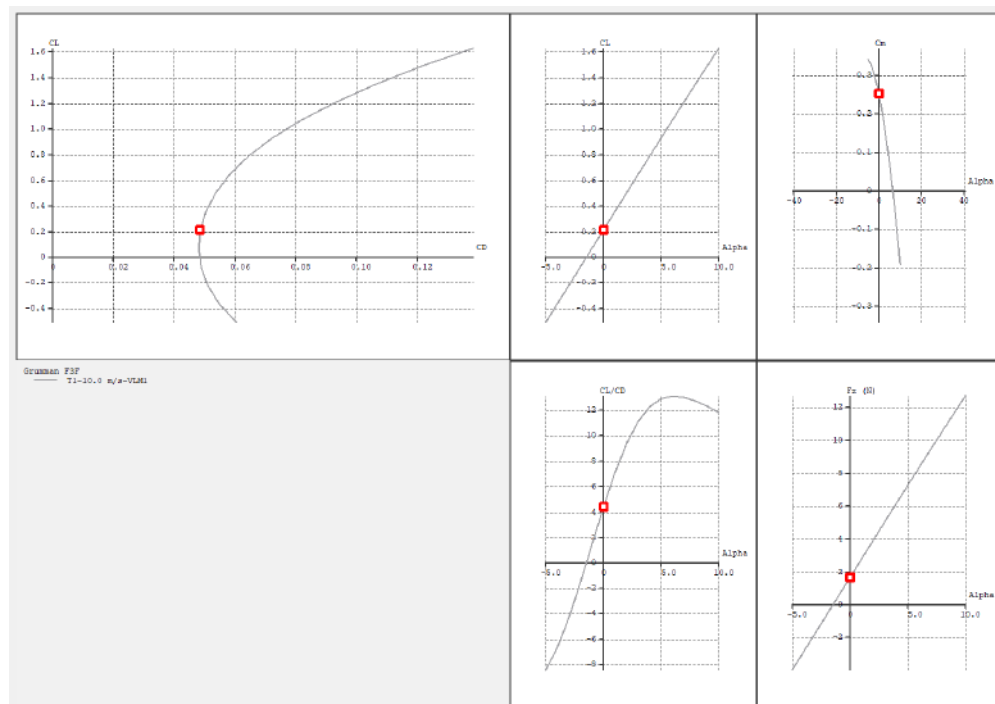


Figure 9: Graphs of Drag polar, lift curve, stability curves, aerodynamic efficiency curve, normal force curve

5 Manufacturing Process

The manufacturing process combined traditional model aircraft building techniques with modern digital fabrication methods:

5.1 CAD Preparation

All parts were designed in CATIA and exported as DXF files for laser cutting. Parts were organized on cutting sheets to minimize material waste. The complex curved sections of the fuselage required particular attention to ensure proper alignment during assembly.

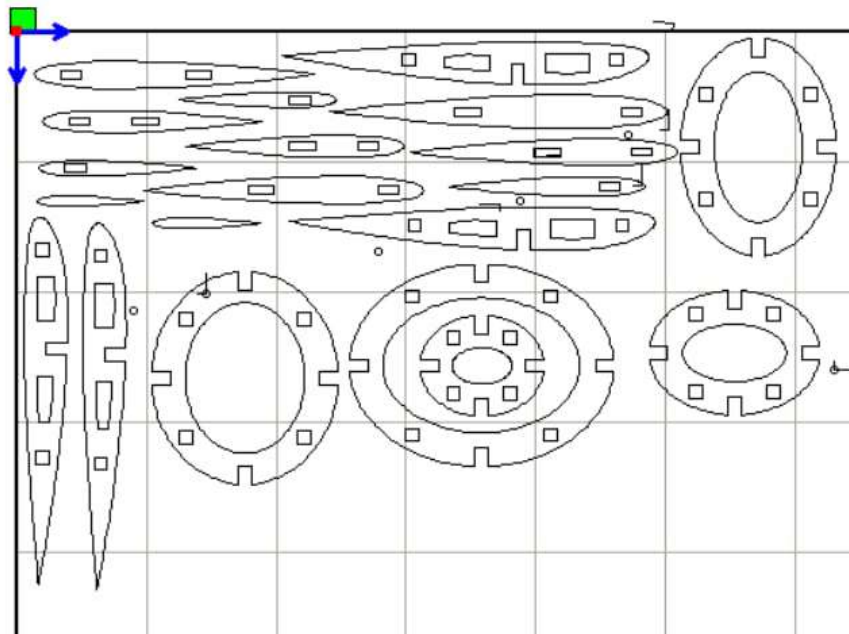


Figure 10: CATIA file in DXF files

5.2 Laser Cutting

Using the laser cutting facility at the manufacturing laboratory, we cut the following components:

- Fuselage formers and stringers
- Wing ribs and spars
- Tail surfaces
- Mounting plates and bracing components

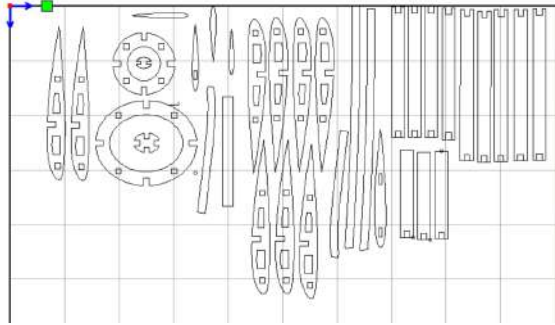


Figure 11: Cut 1

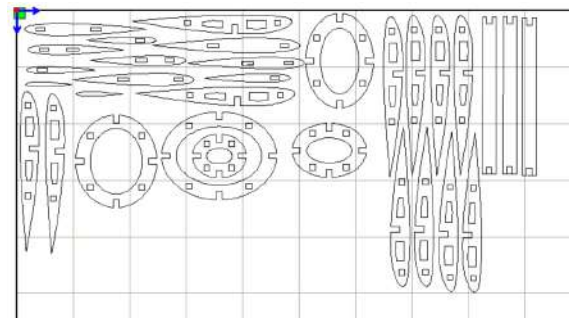


Figure 12: Cut 2

The laser cutting process was conducted in multiple sessions to prevent machine overheating and ensure precision. We maintained focus control throughout the cutting process to achieve clean, accurate cuts in the various thickness materials.

5.3 Assembly Process

1. **Fuselage Assembly:** The fuselage formers were aligned on a building board and connected with stringers to create the characteristic barrel shape. Additional structural reinforcement was added at high stress points.
2. **Wing Assembly:** Wing ribs were threaded onto the main and rear spars, ensuring proper alignment. Leading and trailing edges were added, followed by diagonal bracing between ribs. Each wing was assembled flat on the building board to ensure proper alignment.
3. **Biplane Structure:** The lower wing was attached to the fuselage first, followed by the installation of the N-struts and then the upper wing. Proper alignment of all components was critical to ensure the correct stagger and gap between wings.



Figure 13

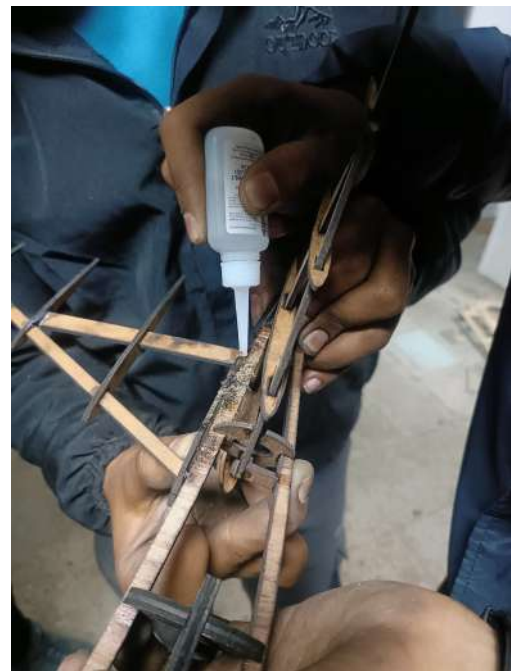


Figure 14



Figure 15

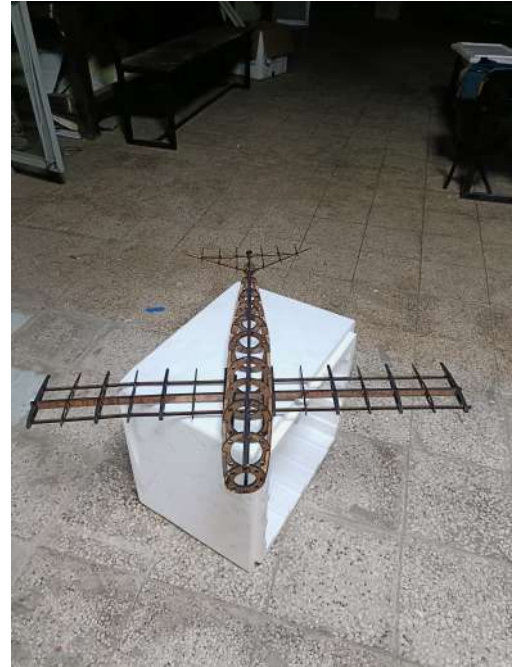


Figure 16

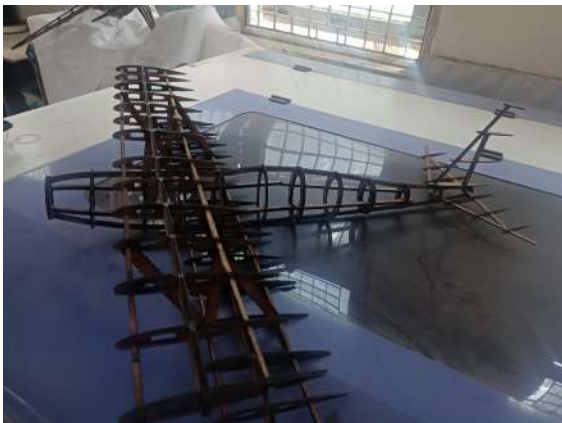


Figure 17

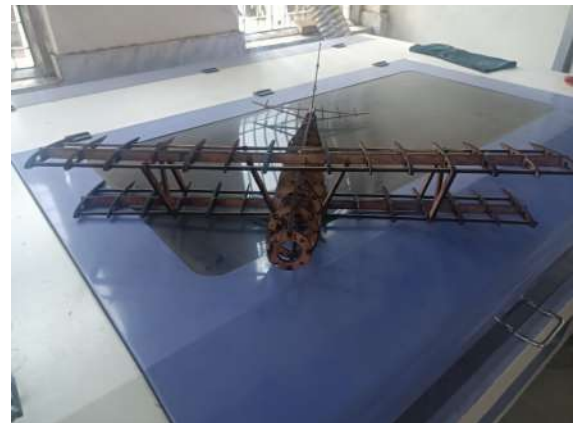


Figure 18

6 Testing

The completed model underwent several tests to verify its structural integrity and aerodynamic performance:

1. **Structural Testing:** Load tests were performed on the wing structure and landing gear to ensure they could withstand flight and landing stresses.
2. **Balance Testing:** The model was balanced at the calculated CG position and adjustments were made as necessary.

7 Conclusion

The 1:10 scale Grumman F3F biplane model successfully captures the distinctive design features and structural properties of this historic aircraft. The biplane configuration presented unique manufacturing challenges, particularly in ensuring proper alignment of the wings and struts. The finished model demonstrates good structural integrity and, according to simulation results, should exhibit stable flight characteristics.

The manufacturing process combined traditional model building techniques with modern digital fabrication methods, allowing for precision parts production while maintaining the craftsmanship necessary for historical aircraft reproduction. This project has enhanced our understanding of aircraft design principles, structural considerations for biplane configurations, and manufacturing techniques for complex aerodynamic structures.